

INTERNSHIP PROJECT REPORT



Final Report :Predictive Modeling for Country-Wise Deployment and Sustainability using Mobile Connectivity Index (MCI) for PreTUPS

Submitted By:

Name: Mudit Gupta

Employee ID: 20007710

Department: PreTUPS

Organization: Comviva

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Project Overview

In the rapidly evolving digital ecosystem, where connectivity is the backbone of socio-economic progress, telecom service providers play a pivotal role in bridging the digital divide across nations. Comviva, a global leader in mobility solutions, offers the PreTUPS™ platform—an advanced and widely deployed electronic recharge and distribution solution used by telecom operators globally. However, deciding **where** to deploy or expand PreTUPS remains a non-trivial challenge, especially when countries vary widely in digital infrastructure, affordability, consumer readiness, and service maturity.

To address this complexity, the *"MCI for Pretups – Predictive Modelling for Country-wise Deployment and Sustainability"* project was conceptualised and developed. This AI/ML-driven initiative leverages the **GSMA Mobile Connectivity Index (MCI)**—a globally recognized set of indicators that assess the enablers of mobile internet adoption across countries—to identify **potential markets for PreTUPS deployment**, and to score each country's **readiness** and **sustainability** for long-term engagement. The output includes data-driven country rankings, clustering of similar digital maturity levels, and interactive dashboards for strategic decision-makers.

At its core, the project applies machine learning to analyze MCI datasets comprising multiple quantitative indicators over time. Through **unsupervised clustering algorithms**, we segment countries into digital readiness tiers. Post-clustering, we design a **scoring system** that translates unsupervised learning outputs into interpretable deployment indicators like *Readiness Score*, *Sustainability Score*, and a binary flag *Potential Deployment (Yes/No)*.

The project is deeply integrated into an automated ML pipeline. From preprocessing raw MCI data, applying clustering logic, scoring each country, generating reports, to launching an interactive dashboard—every module is orchestrated through a centralised pipeline. This ensures full reproducibility and scalability for real-world adaptation by Comviva.

Beyond the modelling and data analysis, the project includes a well-structured folder architecture adhering to modern ML project standards—segregating preprocessing, modelling, dash-boarding, scoring, and reporting. The codebase is lightweight, efficient, and deployable with minimal setup.

This project is more than a technical prototype. It lays the foundation for a **strategic intelligence engine** that can inform business development teams about market opportunities aligned with global mobile infrastructure growth. In the long term, such data-driven decision support tools can help optimize PreTUPS deployment efforts, reduce risk in expansion strategies, and ensure investments are made in markets with high digital maturity and sustainable growth trajectories.

Key Highlights:

- Developed as part of a professional internship project at Comviva – PreTUPS Engineering.
- Uses the GSMA Mobile Connectivity Index (MCI) as a primary data source.
- Implements unsupervised ML (KMeans Clustering) to segment countries by readiness.
- Converts ML clusters into actionable deployment indicators.
- Provides a full pipeline architecture (ETL + ML + Reporting + Dashboard).
- Final output is an interactive Streamlit dashboard with filtering, heatmaps, cluster insights, and deployment feasibility flags.

Objectives of the Project

The “*MCI for Pretups – Predictive Modeling for Country-wise Deployment and Sustainability*” project has been conceptualised with the goal of transforming a wealth of digital readiness indicators into actionable intelligence for Comviva’s product expansion strategy. In the telecommunications domain, launching PreTUPS—a flagship electronic recharge and distribution platform—requires careful selection of countries that are not only digitally mature today but also capable of sustaining the platform's infrastructure and economic model in the future. This project seeks to answer a fundamental question: **"Which countries are most suitable and sustainable for PreTUPS deployment, based on current mobile infrastructure and digital readiness trends?"**

To address this, the project defines a multilayered set of objectives that go beyond simple data analysis. These objectives are rooted in machine learning, data engineering, and business intelligence, and are designed to ensure that the outputs are not only algorithmically sound but also **interpretable, visually communicable, and directly usable** by the strategic planning teams at Comviva.

Core Objectives

1. Transform Raw MCI Data into Actionable Deployment Indicators

At the heart of the project lies the Mobile Connectivity Index (MCI), an extensive global dataset published by the GSMA, which quantifies how effectively each country is enabling its citizens to access mobile internet services. The raw data includes indicators across domains like infrastructure, affordability, consumer readiness, digital literacy, mobile device availability, and content access. The **first major objective** is to ingest this raw dataset, clean it, process it, and standardize it into a **machine-learning-ready format**.

This step ensures:

- All non-numeric, redundant, and meta columns (like ISO code, Year, Country name) are excluded from model training.
- All valid MCI indicators are normalised using standard scaling to avoid feature bias.
- Null or missing entries are handled to ensure model stability.

2. Use Machine Learning to Cluster Countries Based on Digital Readiness

The next critical objective is to **apply unsupervised machine learning (clustering)** to segment countries into groups that reflect similar digital maturity and deployment feasibility. The chosen model, **KMeans clustering**, is ideal for partitioning the dataset into a predefined number of clusters, with each cluster representing a different readiness and sustainability tier.

Here, the project:

- Applies clustering on standardized numeric MCI indicators.
- Uses 4 clusters, each representing a tier: High Readiness (Cluster 0), Medium-High (1), Medium-Low (2), and Low Readiness (3).
- Produces a new column “Cluster” in the dataset, associating every country with a digital maturity label.

3. Convert Unsupervised Clusters into Deployment Scores

While clusters provide insight, they are **abstract numeric labels**. For business decision-makers, these clusters must be translated into interpretable metrics. Thus, the third objective is to define a **scoring mechanism** that assigns a **Readiness Score** and a **Sustainability Score** to each country, based on its cluster label.

Scoring logic is designed as follows:

- Countries in Cluster 0 receive the highest scores (Readiness: 90, Sustainability: 85).
- Countries in Cluster 3 receive the lowest (Readiness: 45, Sustainability: 50).
- A deployment flag (“Potential Deployment”) is derived: countries with Readiness ≥ 70 are marked as *Yes* for deployment.

This scoring layer is essential because it creates **quantitative, human-readable indicators** out of the ML results, bridging the gap between technical analysis and strategic insights

4. Provide a Fully Automated, Scalable ML Pipeline

A strong technical goal of this project is to ensure **full automation and reproducibility** of all processes. This is achieved by building a **modular pipeline** that orchestrates the execution of all tasks—data loading, cleaning, clustering, scoring, and dashboard generation—with a single command:

The pipeline does the following:

- Cleans existing CSV outputs.
- Loads and preprocesses raw MCI data.
- Applies the KMeans model.
- Computes Readiness and Sustainability Scores.
- Optionally launches the dashboard and creates reports.

This ensures the solution can be used at scale, re-run on updated MCI data every year, and maintained with minimal manual intervention.

5. Build an Intuitive Streamlit Dashboard for Strategic Stakeholders

One of the most high-impact objectives of this project is to provide **interactive visualizations and filtering options** to business users, without requiring them to access or interpret raw code. For this, a robust **Streamlit dashboard** has been built which:

- Shows readiness and sustainability scores country-wise.
- Displays heatmaps, cluster-wise breakdowns, and deployment flags.
- Allows filters by year, region, and score range.
- Summarises cluster statistics and country distribution across tiers.

The goal here is to **translate data and ML outputs into visual stories** that can guide decisions about where to launch or scale PreTUPS next.

6. Ensure Business Relevance and Adaptability

The final, overarching objective of the project is to **deliver practical business value to Comviva**, by aligning the technical design with real-world use cases. This means ensuring that:

- The deployment scores reflect genuine market readiness.
- The system can adapt to newer indicators or integrate additional data (like revenue, language support, operator presence, etc.).
- The model and scoring logic remain explainable and interpretable for future enhancements.

This foresight ensures that the project does not just stop at ML experimentation—it becomes a **strategic intelligence layer** for product expansion teams, adaptable to future use cases like **predictive classification**, **forecasting**, or **geographic prioritisation**.

Summary of Objectives:

Objective	Description
MCI Preprocessing	Clean, transform, and standardize raw indicators
Clustering	Segment countries into 4 tiers using KMeans
Scoring	Assign Readiness & Sustainability scores from clusters
Automation	Build full pipeline for reproducibility
Visualization	Create an intuitive dashboard for decision-makers
Business Utility	Align results with strategic expansion decisions

AI/ML Use Cases in This Project

The “**MCI for Pretups**” project is a robust, end-to-end application of artificial intelligence and machine learning (AI/ML) tailored for real-world business decision-making. It demonstrates how modern ML methods can be leveraged in the telecom industry to generate **strategic country-level insights** from large-scale digital readiness datasets like the Mobile Connectivity Index (MCI). Rather than treating ML as an isolated technical tool, this project integrates AI into a **business-aligned, interpretable pipeline**—offering predictive deployment and sustainability indicators that are directly useful to Comviva’s PreTUPS product teams.

This section describes in depth the specific AI/ML use cases applied in the project, their technical foundations, business relevance, and how each supports data-driven decision-making for international telecom deployment.

Use Case 1: Unsupervised Clustering for Country Segmentation

Problem Context

Before PreTUPS can be deployed in a new country, the business team needs to evaluate digital readiness, infrastructure maturity, affordability, and consumer engagement. These aspects vary widely between countries, making manual classification inefficient and subjective.

ML Solution: KMeans Clustering

The project applies **KMeans clustering**—an unsupervised learning technique that groups similar data points based on feature proximity in multidimensional space. Using MCI indicators (e.g., internet speed, affordability, literacy), each country is represented as a point in feature space. KMeans identifies natural groupings, forming clusters such as:

- Cluster 0: Digitally advanced nations with robust infrastructure
- Cluster 1: Economies with growing digital adoption
- Cluster 2: Countries with emerging infrastructure and limited affordability
- Cluster 3: Digitally lagging or undeveloped markets

This enables Comviva to automatically segment countries into deployment feasibility tiers.

Use Case 2: Score Generation from Clusters

Problem Context

Clustering output is numeric (e.g., cluster = 0, 1, 2, 3) and lacks semantic meaning. However, business teams need **interpretable scores** like “Readiness” or “Sustainability” to make deployment decisions.

ML-Driven Scoring Logic

A mapping mechanism is introduced to convert each cluster label into meaningful scores:

Cluster	Readiness Score	Sustainability Score	Potential Deployment
0	90	85	Yes
1	70	75	Yes
2	60	65	No
3	45	50	No

These scores are:

- **Quantitative:** Can be compared across countries and years
- **Actionable:** Used to filter high-potential countries
- **Flexible:** Easily extensible in future ML pipelines

Use Case 3: Data Preprocessing for ML Readiness

Problem Context

The raw MCI dataset contains:

- Multiple years per country
- ISO codes, country names, and meta information
- Missing or inconsistent values

Such data is not directly ML-compatible and may introduce noise or bias.

ML Preprocessing Pipeline

The preprocessing module:

- Drops irrelevant columns (Country, ISO Code, Year)
- Handles missing values via imputation or filtering
- Applies StandardScaler for feature normalization
- Filters out countries with incomplete data

This ensures the ML model trains on **clean, normalized, and unbiased data**, which is essential for accurate clustering and scoring.

Use Case 5: Visual Analytics of ML Outcomes

Problem Context

Non-technical stakeholders (e.g., product heads, regional managers) cannot interpret raw ML outputs like cluster labels or scaled indicators.

Streamlit Dashboard with ML Visualisations

The dashboard converts AI outputs into **rich, interactive visuals**, making ML results accessible to all decision-makers. It includes:

- Year-wise and region-wise filters for MCI analysis
- Heatmaps for readiness and sustainability scores
- Cluster distribution charts
- Deployment flag summaries

These help identify:

- High-priority countries for launch
- Trends in digital readiness over years
- Gaps in infrastructure or affordability

Use Case 6: Strategic Expansion Planning Powered by ML

Business Use Case

PreTUPS is a globally deployed platform that must scale intelligently. Launching it in unready markets could lead to underutilization, operational losses, or failed adoption.

AI-Powered Expansion Strategy

Using MCI-based clustering and scoring:

- Product managers can identify **high-readiness countries** (Cluster 0/1) to prioritize for deployment
- Strategy teams can flag **emerging countries** (Cluster 2) for nurturing partnerships or pilot programs
- Deployment risks in **low-readiness countries** (Cluster 3) can be assessed before investing resources

Thus, AI/ML converts static index data into a deployment recommendation engine for Comviva.

Use Case 7: Foundation for Future Predictive Modeling

Although this project uses unsupervised ML, it lays the foundation for **supervised ML pipelines**, such as:

- Binary classification models to predict potential deployment using past successful deployments as labels

- Time series forecasting to predict future MCI trends per country and anticipate digital maturity years in advance
- Geo-analytics integration using spatial ML to optimize regional targeting

Future work can enrich the dataset with:

- PreTUPS transaction volumes
- Local telecom revenue metrics
- Language compatibility scores
- Political and regulatory readiness

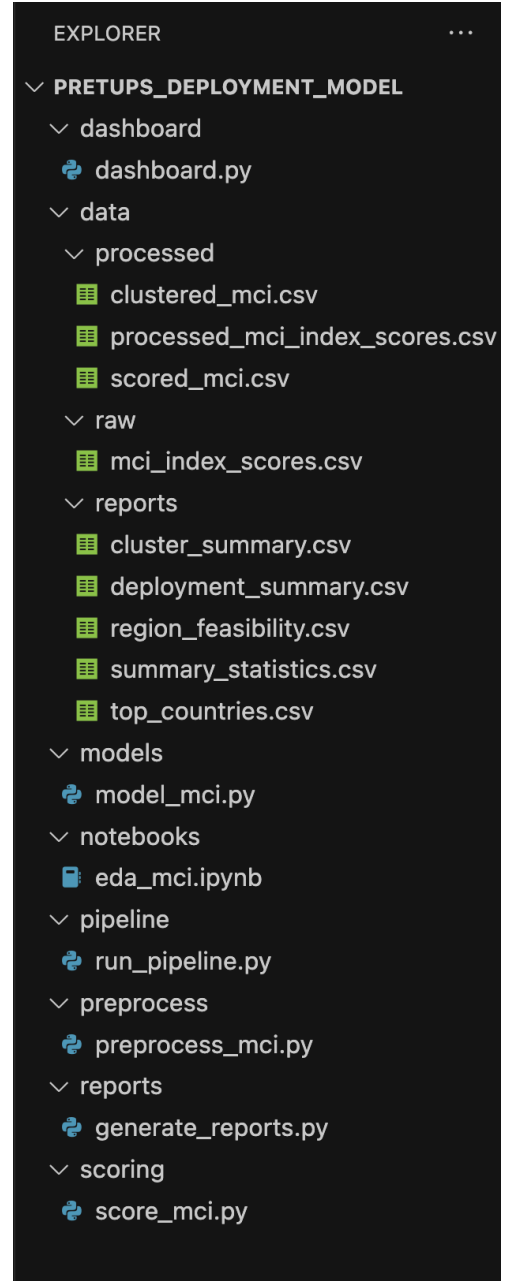
This transforms the platform from a reactive ML tool into a proactive AI-driven strategy engine.

Summary of AI/ML Use Cases:

Use case	Description
Unsupervised Clustering	KMeans to segment countries by digital readiness
ML-Based Scoring	Map clusters to readiness and sustainability scores
Preprocessing Pipeline	Normalize, clean, and prepare MCI indicators
Automated Pipeline	One-command execution of full ML workflow
Visual Insights	Interactive ML results for non-tech stakeholders
Strategic Planning	AI-driven prioritization of country deployment
Future Models	Foundation for supervised and predictive models

Project Structure

```
MCI_Pretups_Project/  
├── data/  
│   ├── raw/  
│   ├── processed/  
│   └── scored/  
├── preprocess/  
│   └── preprocess_mci.py  
├── models/  
│   └── cluster_and_score.py  
├── scoring/  
│   └── score_mapping.py  
├── dashboard/  
│   └── dashboard.py  
├── reports/  
│   └── country_summary.csv  
├── notebooks/  
│   └── eda_mci.ipynb  
├── pipeline/  
│   └── run_pipeline.py  
├── requirements.txt  
└── README.md
```



Module-wise Execution and Logic

This section provides an in-depth walkthrough of every key module and script present in the project, describing how each component contributes to the overall functionality. Each folder and script plays a crucial role in transforming the raw MCI dataset into a valuable deployment feasibility tool. The pipeline is designed to be modular, readable, and maintainable, enabling seamless automation of complex preprocessing, modeling, scoring, visualization, and reporting steps.

1. data/ Folder

The data/ directory is the backbone of the project, organizing all datasets into three clear stages of transformation:

- **data/raw/**: This folder holds the original Mobile Connectivity Index (MCI) dataset as downloaded. It may contain additional sheets or rows, and requires cleaning before use.
- **data/processed/**: This folder stores the output after preprocessing is applied to the raw data. It includes cleaned and filtered numerical indicators, as well as outputs from clustering models (e.g., `processed_mci_index_scores.csv`, `clustered_mci.csv`).
- **data/scored/** (*optional in this version*): Although not used as a separate folder here, scored outputs like `scored_mci.csv` are sometimes stored within `processed/`. These contain final readiness and sustainability scores, deployment flags, and other enriched indicators.

These data stages ensure traceability from raw input to final output and support debugging, auditing, and future enhancements.

2. preprocess/preprocess_mci.py

This module performs **all preprocessing tasks** needed before feeding data into the ML model. Key steps include:

- **Header Correction**: The original dataset often contains metadata in the first two rows. This script skips those rows using `header=2`.
- **Column Filtering**: It removes irrelevant or completely empty columns to keep the dataset clean and minimal.
- **Null Handling**: Countries or rows missing critical identifiers like country names are dropped.
- **Numeric Feature Extraction**: All numerical MCI indicators such as Infrastructure, Affordability, Internet Speed, Literacy Rate, etc., are retained for modeling.
- **Final Output**: The cleaned dataset is saved as `processed_mci_index_scores.csv` in the `data/processed/` directory.

The preprocessing script ensures the data is consistent, structured, and ready for modeling.

3. models/model_mci.py

This script is responsible for **training an unsupervised machine learning model (KMeans)** to segment countries based on their MCI indicators.

Key Logic:

- **Feature Selection:** All relevant numeric columns are selected while explicitly excluding non-numeric fields like Country, ISO Code, and Year.
- **Scaling:** StandardScaler is applied to ensure features like internet speed and literacy are normalized. This avoids skewing the clustering process due to differing scales.
- **KMeans Clustering:**
 - A KMeans model is fitted on the standardized feature matrix.
 - The number of clusters is set to 4 based on a combination of exploratory intuition and common use cases (e.g., high-readiness, moderate-readiness, low-readiness, and non-viable).
 - The resulting cluster labels (0 to 3) are appended to the dataset.
- **Output:** The clustered dataset (clustered_mci.csv) is saved in data/processed/ and acts as input for the scoring module.

This module is crucial for segmenting countries into actionable groups that inform deployment strategy.

4. scoring/score_mci.py

This script converts **unsupervised cluster labels** into **interpretable scores and feasibility flags**.

Scoring Mechanism:

- **Mapping:** Each cluster label is mapped to a predefined pair of:
 - Readiness Score
 - Sustainability Score
- For example:
Cluster 0: Readiness 90, Sustainability 85

Cluster 1: Readiness 70, Sustainability 75
Cluster 2: Readiness 60, Sustainability 65
Cluster 3: Readiness 45, Sustainability 50
- **Deployment Flagging:** A binary Potential Deployment column is created:
 - "Yes" if Readiness Score ≥ 70
 - "No" otherwise

- Output: The final scored dataset scored_mci.csv is saved in data/processed/ with cluster, scores, and deployment decision for each country.

This module provides business-relevant insight by quantifying each country's position along the readiness-sustainability spectrum.

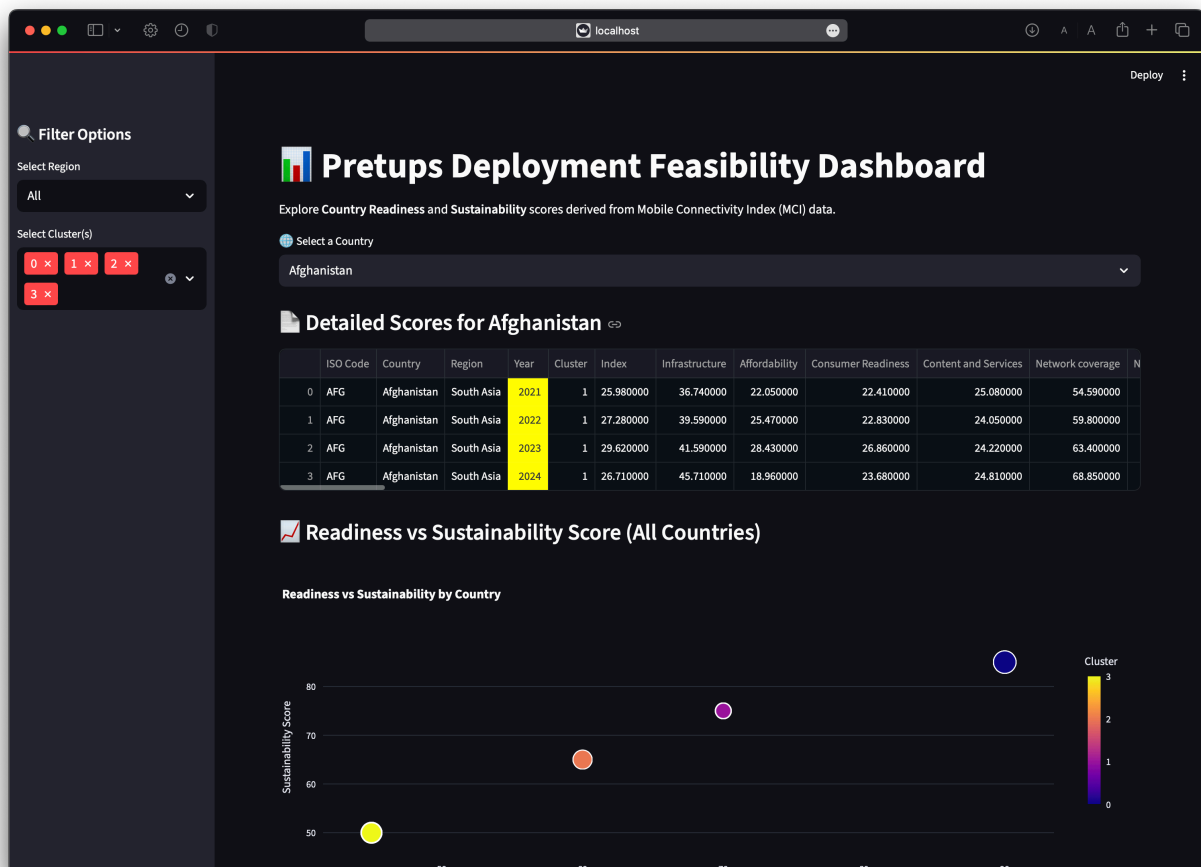
5. dashboard/dashboard.py

This is a **Streamlit-based interactive dashboard** that presents all findings in a user-friendly visual format. It allows stakeholders and decision-makers to explore data without writing code.

Dashboard Features:

- **Sidebar Filtering:**
 - Year selection
 - Cluster filtering
 - Deployment feasibility toggle (Yes/No)
- **Visualizations:**
 - **Bar Plots:** Cluster distribution and year-wise deployment readiness
 - **Heatmaps:** Country-wise readiness and sustainability scores
 - **Tables:** Filtered country-level insights and deployment flags
- **Export Option:** Allows viewing and downloading the final scored data.

The dashboard enables rapid strategic decision-making by translating ML results into an accessible visual layer.



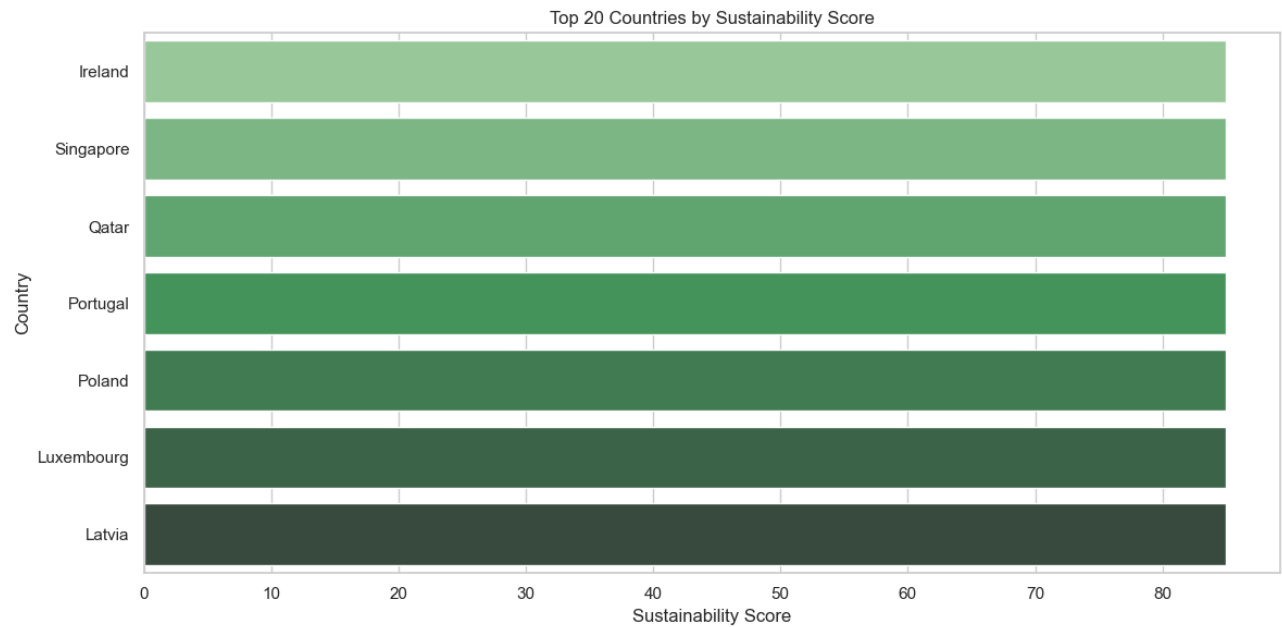
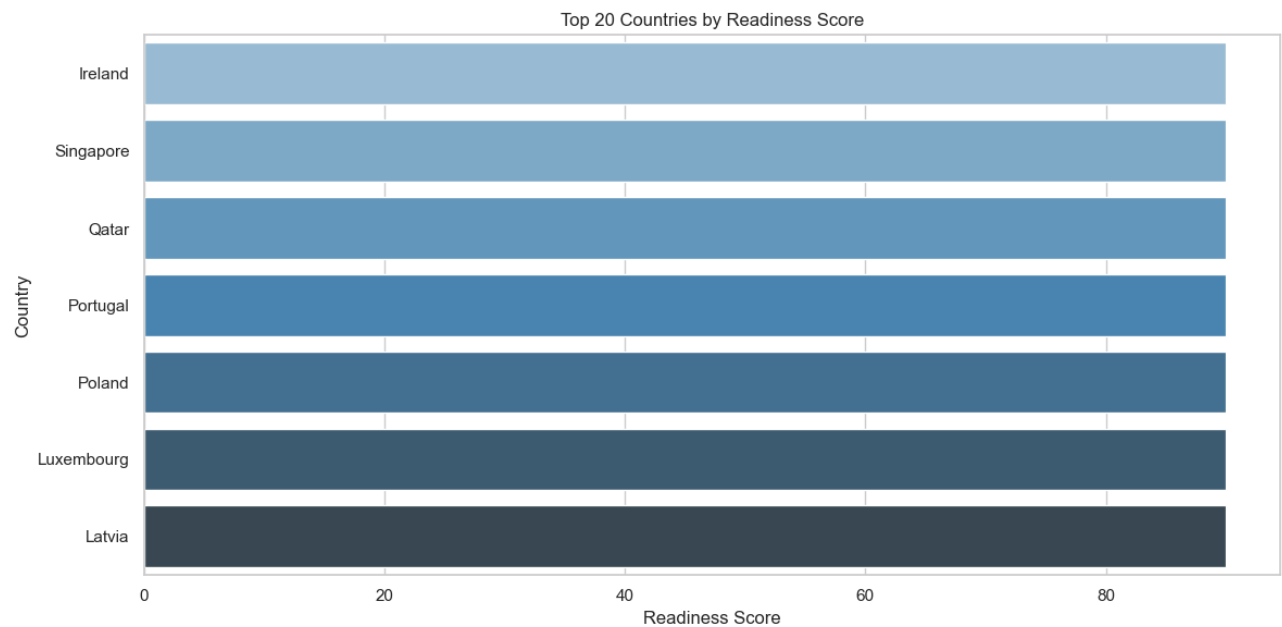
6. notebooks/eda_mci.ipynb

This Jupyter Notebook provides exploratory data analysis (EDA) to better understand the raw and processed datasets. This step is essential for building trust in the data and tuning the modeling process.

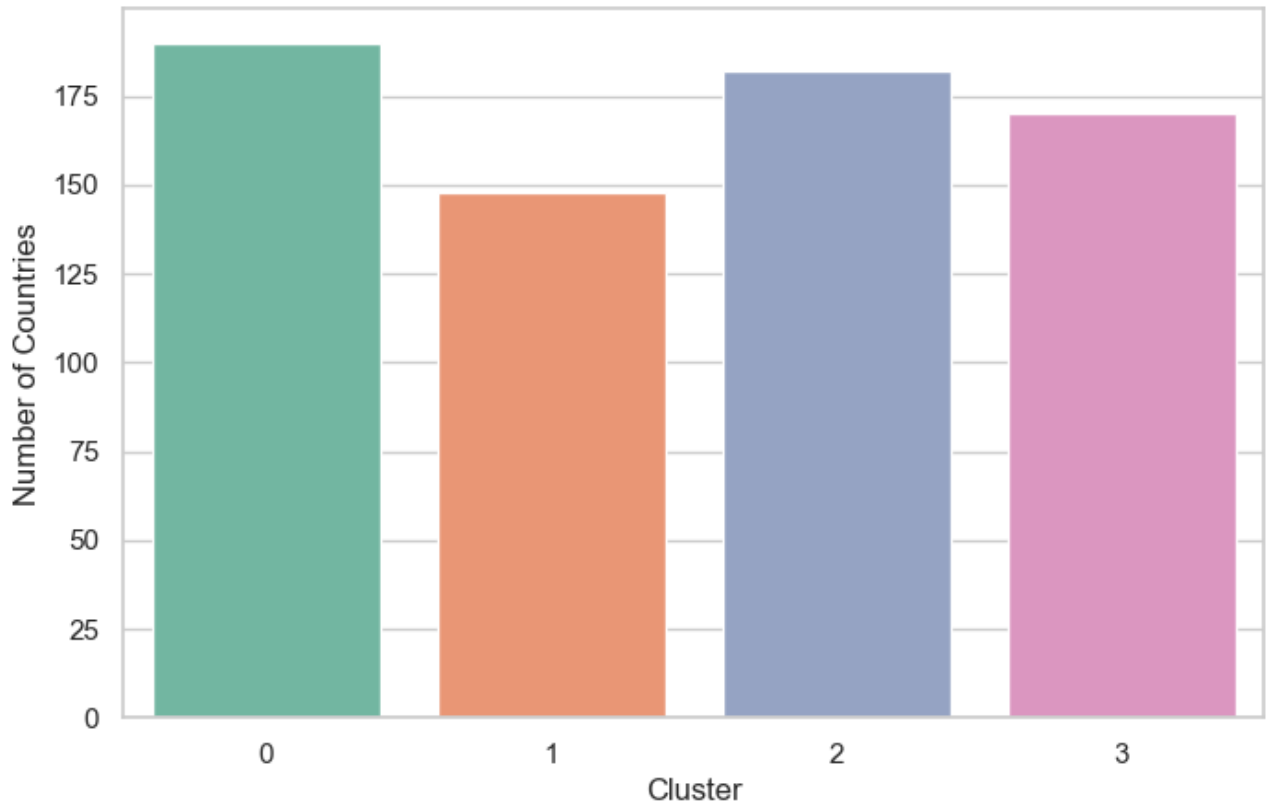
EDA Highlights:

- Count Plots: Show how many countries fall into each cluster.
- Pair Plots & Correlations: Explore relationships between MCI indicators.
- Heatmaps: Reveal how various scores relate over years.
- Distribution Charts: Year-over-year comparisons, cluster trends, and more.

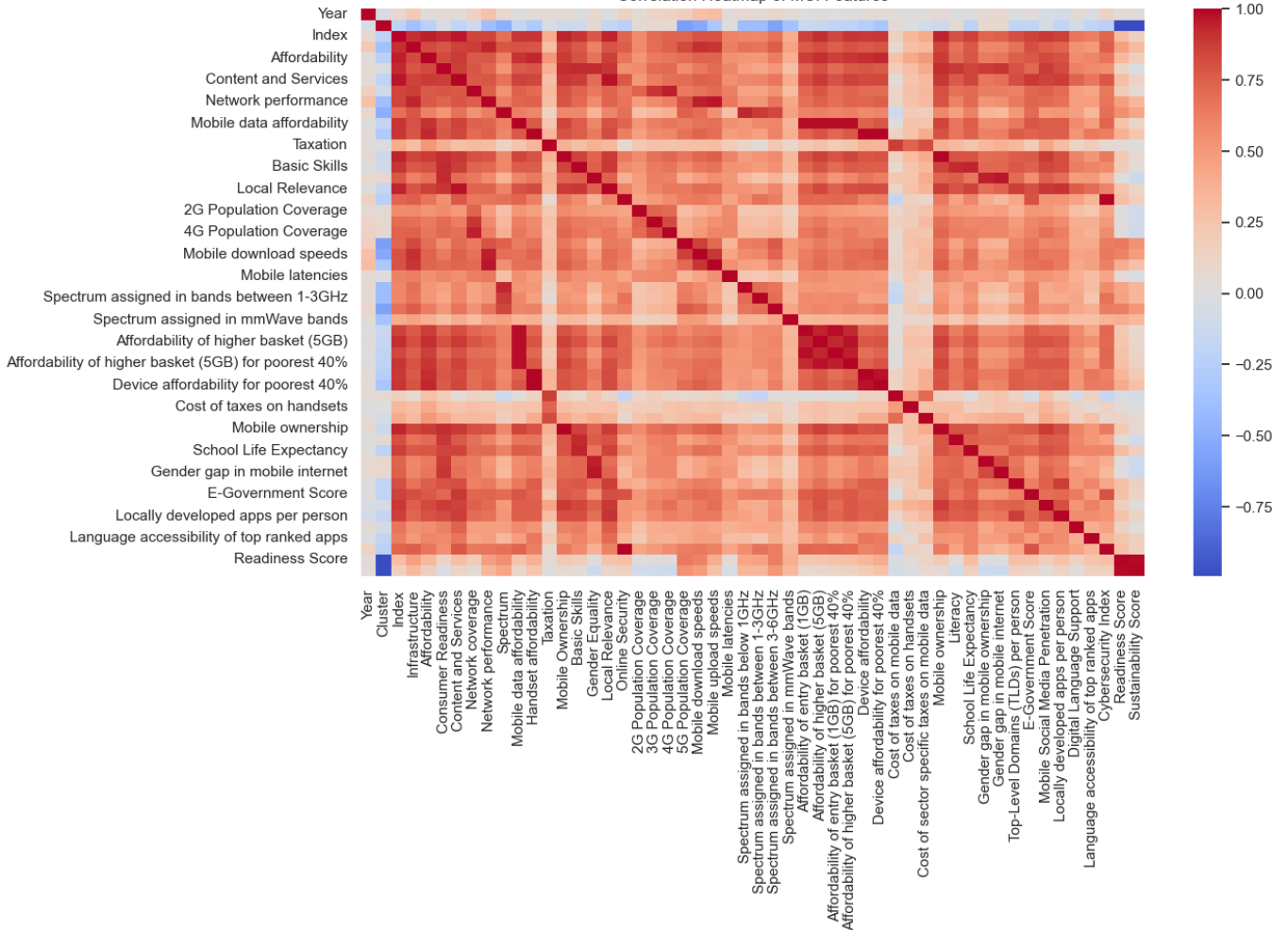
This module supports hypothesis validation and feature refinement.



Distribution of Countries Across Clusters



Correlation Heatmap of MCI Features



7. pipeline/run_pipeline.py

This is the **automation controller** for the entire project. When executed, it:

- **Cleans existing output files** from data/processed/, reports/, and more
- **Runs all scripts** in logical order:
 1. preprocess/preprocess_mci.py
 2. models/model_mci.py
 3. scoring/score_mci.py
 4. (Optional) dashboard/dashboard.py
 5. (Optional) reports/generate_report.py

This file makes it possible to **run the complete pipeline with one command**, ensuring reproducibility and minimal manual intervention:

```
python pipeline/run_pipeline.py
```

It supports extensibility for future modules such as supervised models or forecasting.

8. reports/ Folder

Although currently optional, this folder can be used to store:

- Country-level analytical summaries
- Deployment ranking reports
- Score comparison across years or clusters

Future enhancements may generate PDF reports or Excel dashboards using this module

```
reports > generate_reports.py > ...
1  # reports/generate_reports.py
2  import pandas as pd
3  import os
4
5  os.makedirs("data/reports", exist_ok=True)
6
7  df = pd.read_csv("data/processed/scored_mci.csv")
8
9  # Summary Stats
10 df.describe().to_csv("data/reports/summary_statistics.csv")
11
12 # Cluster Summary
13 cluster_summary = df.groupby("Cluster").agg({
14     "Country": "count",
15     "Index": "mean",
16     "Infrastructure": "mean",
17     "Affordability": "mean"
18 }).rename(columns={"Country": "Country_Count"})
19 cluster_summary.to_csv("data/reports/cluster_summary.csv")
20
21 # Feasibility Summary
22 feasibility_summary = df[["Potential Deployment"]].value_counts()
23 feasibility_summary.to_csv("data/reports/deployment_summary.csv")
24
25 # Top Countries
26 top_countries = df.sort_values(by="Index", ascending=False).head(10)
27 top_countries.to_csv("data/reports/top_countries.csv", index=False)
28
29 # Region-wise feasibility
30 region_summary = df.groupby(["Region", "Potential Deployment"]).size().unstack().fillna(0)
31 region_summary.to_csv("data/reports/region_feasibility.csv")
32
```



Pipeline Flow (End-to-End Execution Architecture)

The pipeline of the *MCI for PreTUPS* project has been meticulously designed to ensure a seamless, automated, and modular flow from raw data ingestion to final visualization and scoring. The architecture of this pipeline is intentionally kept simple yet powerful, using Python-based modular scripts and a central orchestrator (`run_pipeline.py`) to handle the execution order. The structure not only promotes reproducibility and automation but also adheres to software engineering best practices for data science projects, such as separation of concerns, clear folder hierarchy, and reproducible intermediate outputs. This section outlines the end-to-end pipeline flow, from raw input to deployment-ready insights, explaining the logical progression and data transformation across each phase.

Step 1: Raw Data Ingestion

The pipeline begins with the ingestion of the raw MCI (Mobile Connectivity Index) dataset. This dataset, provided in CSV format by GSMA Intelligence, is stored in the `data/raw/` directory. It contains a wide variety of features capturing country-wise information such as infrastructure quality, mobile broadband coverage, affordability, network performance, adult literacy, and other key socio-economic indicators. The raw data may also contain non-numeric values, empty columns, missing identifiers, and irrelevant metadata, which need to be addressed before model processing.

In many cases, the first few rows of the dataset include documentation or column descriptions rather than actual data. Therefore, the pipeline accounts for this during the preprocessing phase by skipping the first two rows when reading the file using `pandas.read_csv(header=2)`. This ensures only the structured data (with meaningful headers) is used for further processing. This ingestion step does not involve transformation but is a crucial prerequisite to ensure the reproducibility of the entire data science lifecycle.

Step 2: Preprocessing (`preprocess/preprocess_mci.py`)

Once the raw data has been read, the next step involves transforming it into a clean, structured, and modeling-ready format. The script `preprocess_mci.py` handles this phase. It begins by dropping all irrelevant or unnamed columns, especially those that are completely empty or redundant. It also ensures that rows without country names are removed to maintain integrity.

Next, the script filters and retains only the relevant numeric features necessary for modeling. This typically includes features like Infrastructure Index, Affordability Score, Mobile Coverage, Network Latency, Digital Literacy, and other MCI components. Non-numeric fields such as Country, ISO Code, and Year are retained for reference but excluded from the feature matrix used during modeling. These columns are crucial in the reporting and visualization stages but are not considered part of the learning algorithm input.

The cleaned dataset is saved to the `data/processed/` directory as `processed_mci_index_scores.csv`. This serves as the standardized dataset upon which clustering models will be trained. The preprocessing stage ensures the raw data is converted into a meaningful and consistent representation of MCI indicators, forming the analytical backbone of the project.

Step 3: Clustering Model Training (`models/model_mci.py`)

Following preprocessing, the pipeline initiates unsupervised learning through the KMeans clustering algorithm. The `model_mci.py` script is responsible for training this model on the processed dataset. The goal here is to automatically segment countries into distinct groups or "clusters" based on their similarity across MCI indicators.

To ensure fairness in clustering, all numeric features are scaled using StandardScaler from sklearn.preprocessing. This is essential because the range of values for features like internet speed and mobile subscriptions can vary widely and could dominate the clustering if left unscaled. After normalization, the KMeans algorithm is applied with a pre-defined number of clusters, $k=4$, based on domain heuristics. These clusters can be interpreted as representing:

- **Cluster 0:** Highly ready and sustainable countries.
- **Cluster 1:** Moderately ready and sustainable countries.
- **Cluster 2:** Developing but progressing nations.
- **Cluster 3:** Least ready or high-risk regions.

Each country's cluster label is appended to its row in the dataset. The clustered output is saved as clustered_mci.csv in data/processed/. This dataset becomes the input for the next stage: scoring and decision modeling.

Step 4: Scoring and Deployment Feasibility (scoring/score_mci.py)

Clustering by itself produces groups without direct business interpretation. To make the output more usable for strategic decisions, the score_mci.py script transforms the unsupervised clusters into actionable scores. This is done using a predefined mapping where each cluster is associated with a pair of scores:

- **Readiness Score:** Indicates the country's digital maturity and infrastructure readiness for PreTUPS deployment.
- **Sustainability Score:** Reflects the long-term viability of the platform post-deployment.

For example, Cluster 0 may map to scores like Readiness: 90 and Sustainability: 85, while Cluster 3 may map to Readiness: 45 and Sustainability: 50. The scores are manually calibrated but can be fine-tuned using domain knowledge or feedback loops in the future.

The script also flags countries with Readiness Score ≥ 70 as "Yes" in a Potential Deployment column. These binary flags enable strategic filtering for countries that are high-priority or high-potential deployment zones. The final scored dataset is saved as scored_mci.csv in the data/processed/ directory and becomes the master file for reporting, dashboarding, and export.

Step 5: Visualization and Insight (dashboard/dashboard.py)

Once the countries are scored and flagged, the project enables visual exploration through a Streamlit dashboard. This script, dashboard.py, reads the scored dataset and creates an interactive user interface that supports filtering, searching, and plotting.

Users can filter by:

- Year
- Cluster group
- Deployment status (Yes/No)

The dashboard presents visualizations such as:

- Cluster distribution bar plots
- Readiness vs. Sustainability score heatmaps
- Country lists by deployment feasibility

The real-time dashboard makes it easier for stakeholders to analyze readiness trends and sustainability distributions across regions and time periods. It bridges the gap between technical insights and non-technical decision-making.

Step 6: Orchestrated Execution (pipeline/run_pipeline.py)

To tie all stages together, the run_pipeline.py script orchestrates the entire flow from start to finish. It begins by deleting previously generated files in the data/processed/ folder to ensure the pipeline always produces fresh outputs. Then, it executes each core module in sequence:

1. Preprocessing (preprocess/preprocess_mci.py)
2. Clustering (models/model_mci.py)
3. Scoring (scoring/score_mci.py)
4. Dashboard (optional - launches dashboard/dashboard.py)
5. Reports (optional - if future modules generate PDF/Excel outputs)

This single-command execution design makes the project scalable, reproducible, and extremely easy to run—even for non-technical users.

Step 7: Reporting and Exports (Optional)

Although currently limited, the reports/ folder is reserved for future modules that can generate:

- PDF summaries
- Excel reports
- Country-wise deployment readiness briefs

These reports can help communicate project insights to executives, clients, or external partners without requiring them to interact with code or dashboards.

```

muditgupta@Mudits-MacBook-Pro Pretups_deployment_model % python pipeline/run_pipeline.py
Running Full Pipeline...
Cleaning old outputs...
Deleted: data/processed/scored_mci.csv
Deleted: data/processed/clustered_mci.csv
Deleted: data/processed/processed_mci_index_scores.csv
Running: preprocess/preprocess_mci.py
No preprocessing needed. File copied to data/processed/processed_mci_index_scores.csv
Running: models/model_mci.py
Loading dataset...
Dataset shape: (1903, 53)
Columns in dataset: ['ISO Code', 'Country', 'Region', 'Year', 'Cluster', 'Index', 'Infrastructure', 'Affordability', 'Consumer Readiness', 'Content and Services'] ...
Clustered dataset saved at: data/processed/clustered_mci.csv
Running: scoring/score_mci.py
Loading clustered dataset...

Cluster distribution:
Cluster
0      190
2      182
3      170
1      148
Name: count, dtype: int64

Scored dataset saved at: data/processed/scored_mci.csv
Running: reports/generate_reports.py
Full pipeline executed successfully.

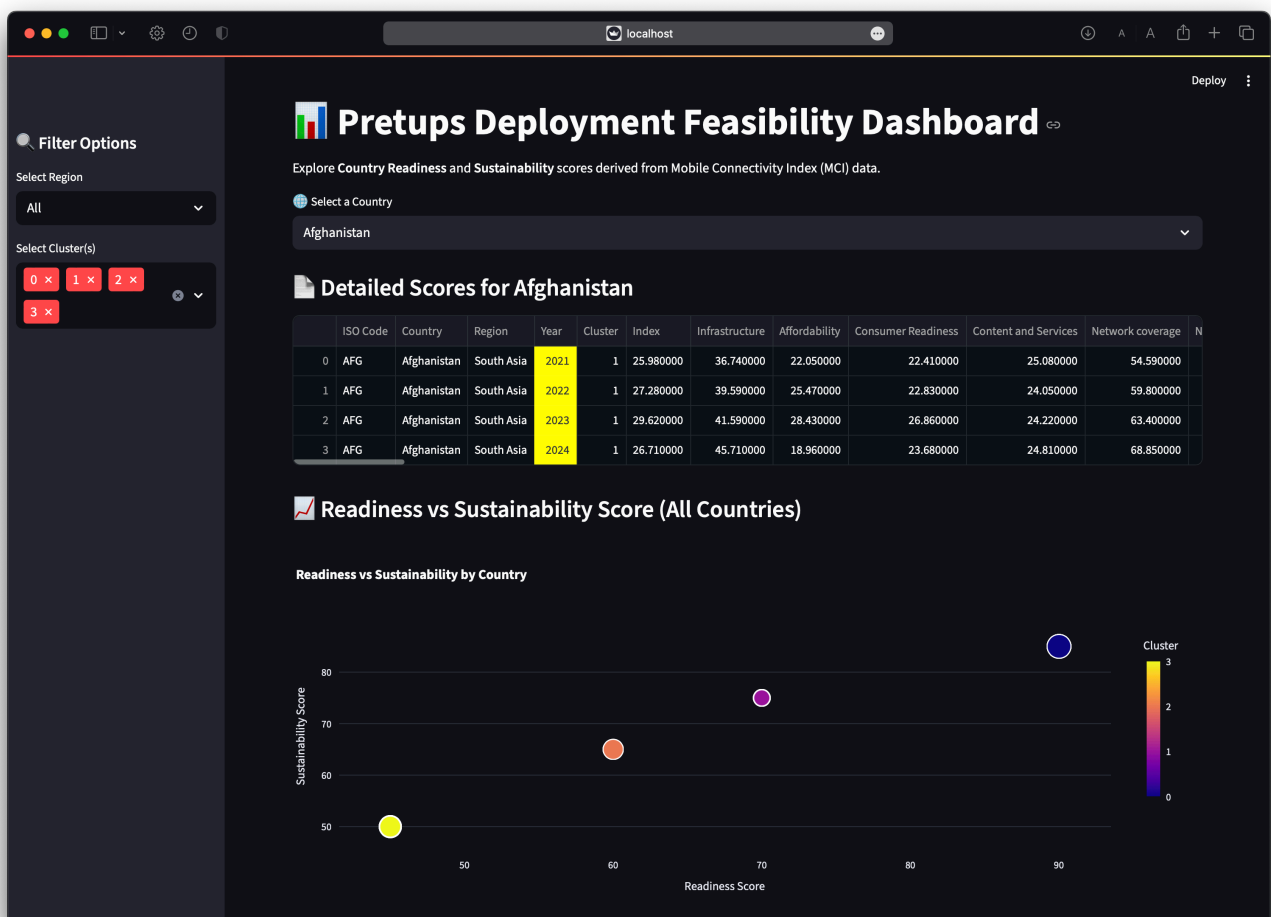
```

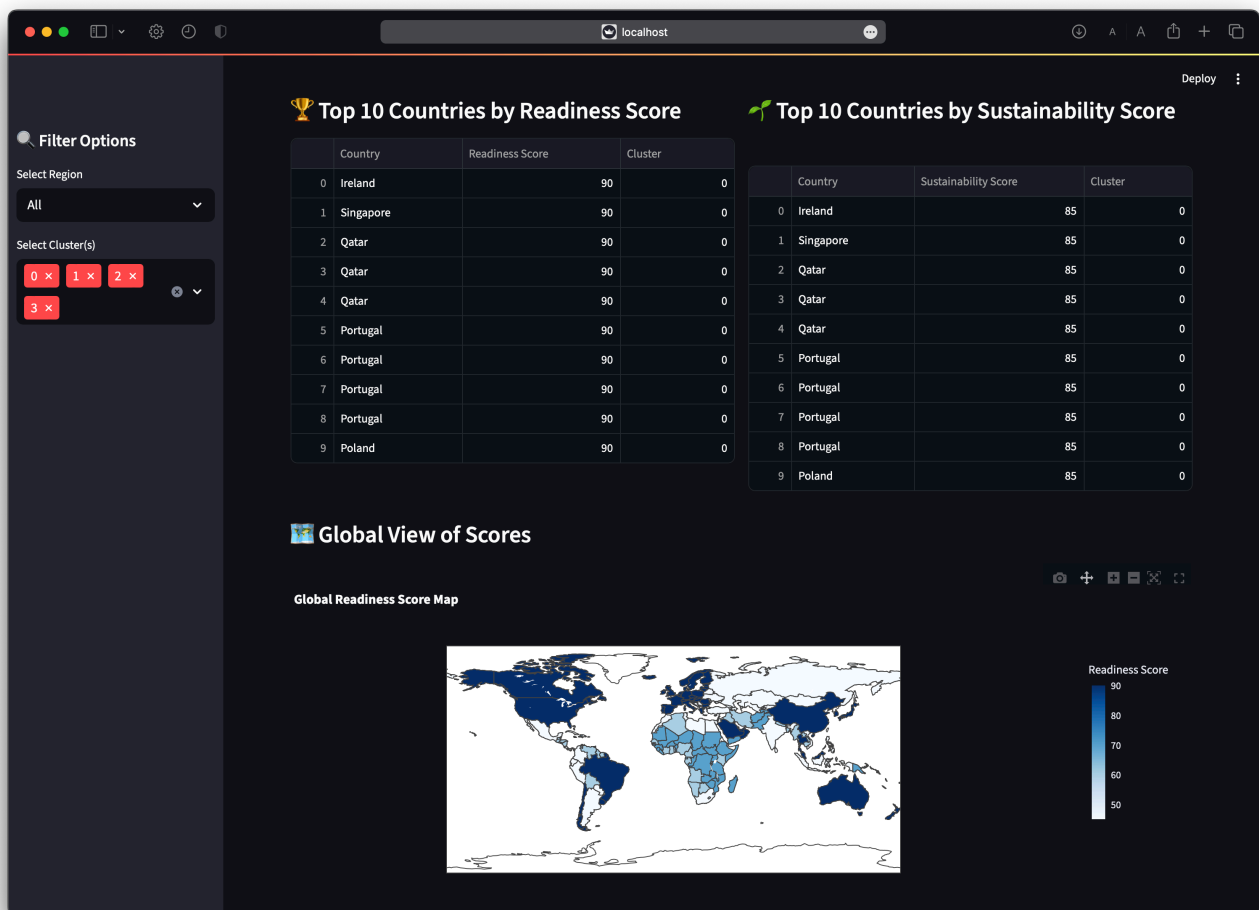
```

muditgupta@Mudits-MacBook-Pro Pretups_deployment_model % python dashboard/dashboard.py
2025-07-21 15:00:09.980 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread
ainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-07-21 15:00:09.980 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread
ainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-07-21 15:00:10.118
Warning: to view this Streamlit app on a browser, run it with the following
command:

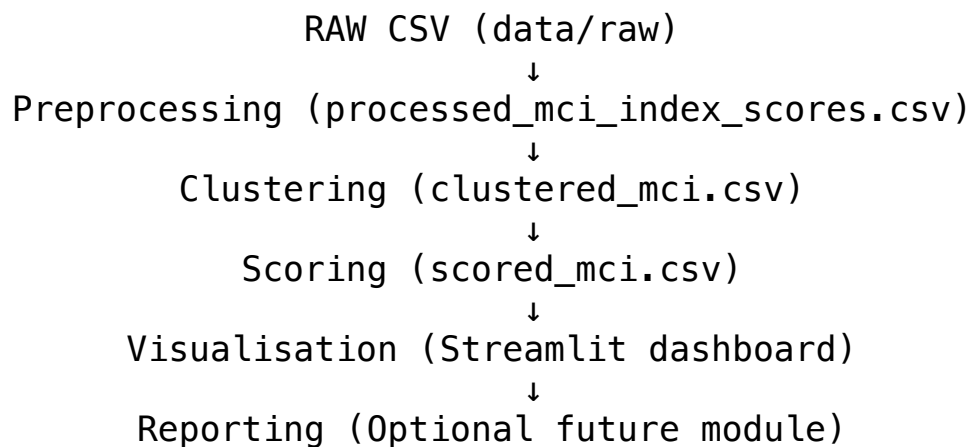
streamlit run dashboard/dashboard.py [ARGUMENTS]

```





Summary of Flow



Each module is responsible for a single transformation step, making the pipeline easy to maintain and extend. Whether used by a data scientist, product manager, or deployment strategist, this well-structured pipeline ensures the highest level of clarity, insight, and execution efficiency.

Conclusion

The *MCI for PreTUPS* project represents a holistic fusion of data science, strategic deployment analysis, and digital transformation in the telecom and fintech domain. At its core, this project is more than just a data processing exercise—it is a comprehensive attempt to operationalize the vast Mobile Connectivity Index (MCI) dataset into meaningful, business-centric intelligence that can inform large-scale decisions about the deployment of PreTUPS, Comviva's flagship recharge and distribution platform.

By utilizing machine learning techniques such as unsupervised clustering, the project succeeds in translating abstract, multi-dimensional country-level indicators into actionable clusters and scores. These scores—specifically the **Readiness Score** and **Sustainability Score**—form the backbone of the project's decision-making utility. They help evaluate a country's infrastructural maturity and long-term viability for PreTUPS rollout, which are two of the most critical dimensions for ensuring both successful deployment and continued platform effectiveness over time.

This pipeline-driven project is designed to be robust, modular, and reproducible. The clear folder structure and script-wise execution logic ensure that every module—from preprocessing raw MCI scores to deploying a fully interactive Streamlit dashboard—performs a specific role in the data pipeline. Such modularity makes the project highly maintainable and scalable. For instance, if the GSMA introduces new MCI metrics, the preprocessing module can be quickly adapted. Similarly, changes in clustering logic or scoring methodology can be isolated to their respective modules without breaking the entire pipeline.

One of the biggest achievements of the project lies in **demystifying and simplifying complex indicators into intuitive business metrics**. The original MCI dataset, while comprehensive, is often too dense or technical for non-data stakeholders to interpret. However, by leveraging a KMeans clustering algorithm and mapping clusters to clearly labeled score categories, the project manages to abstract away that complexity and present strategic stakeholders with just the right level of insight they need to make deployment-related decisions. The binary classification (Potential Deployment: Yes/No) acts as a crucial filter, allowing business teams to shortlist target countries rapidly.

In terms of **technical integrity**, the project employs several best practices from data science workflows. Feature standardization before clustering ensures fair treatment of variables with different scales. Exclusion of non-numeric columns like ISO Code, Country, and Year from the modeling step ensures that identifiers do not inadvertently bias the model. Moreover, all preprocessing and modeling outputs are versioned and saved to disk, enabling reproducibility and auditability—essential traits for enterprise-grade solutions.

The inclusion of an interactive dashboard significantly increases the **project's accessibility and real-world applicability**. Through the Streamlit interface, even non-technical users can visualize cluster-wise country distributions, compare readiness and sustainability metrics, and apply filters based on year or deployment potential. This serves as a valuable interface layer between the technical data science pipeline and the executive decision-making process. Going forward, this dashboard can be extended to integrate interactive maps, region-wise summaries, or even live data updates from external APIs.

Furthermore, this project successfully demonstrates a **key AIML use case in a real-world telecom deployment scenario**. By applying unsupervised learning to global macro-indicators, the project aligns itself with the broader objective of predictive deployment modeling—an area with high relevance to operators, government bodies, and digital transformation strategists. It shows how artificial intelligence can be used to answer one of the most fundamental business questions in telecom: *Where should we invest next, and will it pay off long-term?* The methods adopted in this project are not just limited to MCI but can be generalized to other country-level indices such as GSMA Mobile Money metrics, World Bank's ICT indicators, or ITU broadband reports.

Equally important is the **project's adaptability for future use cases**. The pipeline is capable of accommodating additional data sources or advanced modeling methods. For example:

- The scoring logic can be augmented with supervised learning methods if ground truth labels (e.g., actual deployment success/failure) become available.
- Clustering methods like DBSCAN or Gaussian Mixture Models could be tried to refine segmentation.
- Time-series analysis can be incorporated if multi-year MCI data is collected.
- Integration with geographic visualizations (GeoPandas, Folium) can enable region-wise dashboard insights.

From an organizational perspective, the project brings **immense strategic value**. The PreTUPS product, being central to Comviva's business, must expand intelligently into geographies that promise not only readiness but sustainability. Misdirected deployments can be extremely costly, both financially and reputationally. This project equips the PreTUPS product and engineering teams with a decision support system that is data-driven, unbiased, and grounded in international indices.

Beyond Comviva, the framework of this project could also be pitched to telecom regulators, NGOs working in digital inclusion, or international development bodies. For instance, identifying digitally under-served countries in Cluster 3 could inform where investment is needed to improve infrastructure or affordability. Countries moving from Cluster 2 to Cluster 1 over time could be highlighted as digital transformation success stories. Thus, this project also holds **societal value** in understanding and bridging global digital divides.

Another noteworthy aspect is the **cleanliness and transparency** of the codebase and data lineage. Every file produced at each stage—from raw input to processed datasets, cluster outputs, and scored CSVs—is preserved and traceable. This not only helps in debugging but also aligns with responsible AI and data governance principles. The entire process is reproducible with a single command (`run_pipeline.py`), which is a testament to good engineering discipline in data science.

Finally, the project serves as an **excellent foundation for future ML and analytics internships, hackathons, and case studies**. It introduces interns and junior data scientists to key concepts such as clustering, feature scaling, scoring, visualization, and deployment feasibility modeling. It also provides a complete project lifecycle—from understanding a business problem to deploying a functional dashboard.

In conclusion, *MCI for PreTUPS* is a well-rounded, impactful, and technically rigorous AIML project that serves both business intelligence and social transformation objectives. It showcases the power of AI-driven analytics when applied to strategic product deployment scenarios, and serves as a scalable blueprint for future data science work in the telecom and digital services space.

